

Newman-Penrose Formalism Calculations

Stephani et al., Equation (22.53)

Metric

$$ds^2 = -e^{(-2U+2k)} dt \otimes dt + e^{(-2U+2k)} d\rho \otimes d\rho + W^2 e^{(-2U)} d\varphi \otimes d\varphi + e^{(2U)} dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, \rho, \varphi, z)$$

Metric Functions

$$W(t, \rho) = W$$

$$k(t, \rho) = k$$

$$U(t, \rho) = U$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} e^{(-U+k)} \right) dt + \left(\frac{1}{2} \sqrt{2} e^{(-U+k)} \right) d\rho$$

$$n = \left(\frac{1}{2} \sqrt{2} e^{(-U+k)} \right) dt + \left(-\frac{1}{2} \sqrt{2} e^{(-U+k)} \right) d\rho$$

$$m = \left(\frac{1}{2} \sqrt{2} W e^{(-U)} \right) d\varphi + \left(\frac{1}{2} i \sqrt{2} e^U \right) dz$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} W e^{(-U)} \right) d\varphi + \left(-\frac{1}{2} i \sqrt{2} e^U \right) dz$$

Contravariant components

$$l = \left(-\frac{1}{2} \sqrt{2} e^{(U-k)} \right) dt + \left(\frac{1}{2} \sqrt{2} e^{(U-k)} \right) d\rho$$

$$n = \left(-\frac{1}{2} \sqrt{2} e^{(U-k)} \right) dt + \left(-\frac{1}{2} \sqrt{2} e^{(U-k)} \right) d\rho$$

$$m = \left(\frac{\sqrt{2} e^U}{2W} \right) d\varphi + \left(\frac{1}{2} i \sqrt{2} e^{(-U)} \right) dz$$

$$\bar{m} = \left(\frac{\sqrt{2} e^U}{2W} \right) d\varphi + \left(-\frac{1}{2} i \sqrt{2} e^{(-U)} \right) dz$$

Directional Derivatives

$$D = -\frac{1}{2} \sqrt{2} e^{(U-k)} \partial_t + \frac{1}{2} \sqrt{2} e^{(U-k)} \partial_\rho$$

$$\Delta = -\frac{1}{2} \sqrt{2} e^{(U-k)} \partial_t - \frac{1}{2} \sqrt{2} e^{(U-k)} \partial_\rho$$

$$\delta = \frac{\sqrt{2} e^U}{2W} \partial_\varphi + \frac{1}{2} i \sqrt{2} e^{(-U)} \partial_z$$

$$\bar{\delta} = \frac{\sqrt{2} e^U}{2W} \partial_\varphi - \frac{1}{2} i \sqrt{2} e^{(-U)} \partial_z$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}e^{(U-k)}\frac{\partial}{\partial t}W}{4W} - \frac{\sqrt{2}e^{(U-k)}\frac{\partial}{\partial \rho}W}{4W}$$

$$\sigma = -\frac{1}{2}\sqrt{2}e^{(U-k)}\frac{\partial}{\partial t}U + \frac{1}{2}\sqrt{2}e^{(U-k)}\frac{\partial}{\partial \rho}U + \frac{\sqrt{2}e^{(U-k)}\frac{\partial}{\partial t}W}{4W} - \frac{\sqrt{2}e^{(U-k)}\frac{\partial}{\partial \rho}W}{4W}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}e^{(U-k)}\frac{\partial}{\partial t}W}{4W} - \frac{\sqrt{2}e^{(U-k)}\frac{\partial}{\partial \rho}W}{4W}$$

$$\lambda = \frac{1}{2}\sqrt{2}e^{(U-k)}\frac{\partial}{\partial t}U + \frac{1}{2}\sqrt{2}e^{(U-k)}\frac{\partial}{\partial \rho}U - \frac{\sqrt{2}e^{(U-k)}\frac{\partial}{\partial t}W}{4W} - \frac{\sqrt{2}e^{(U-k)}\frac{\partial}{\partial \rho}W}{4W}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{1}{4}\sqrt{2}e^{(U-k)}\frac{\partial}{\partial t}U - \frac{1}{4}\sqrt{2}e^{(U-k)}\frac{\partial}{\partial \rho}U - \frac{1}{4}\sqrt{2}e^{(U-k)}\frac{\partial}{\partial t}k + \frac{1}{4}\sqrt{2}e^{(U-k)}\frac{\partial}{\partial \rho}k$$

$$\gamma = -\frac{1}{4}\sqrt{2}e^{(U-k)}\frac{\partial}{\partial t}U - \frac{1}{4}\sqrt{2}e^{(U-k)}\frac{\partial}{\partial \rho}U + \frac{1}{4}\sqrt{2}e^{(U-k)}\frac{\partial}{\partial t}k + \frac{1}{4}\sqrt{2}e^{(U-k)}\frac{\partial}{\partial \rho}k$$

Ricci Tensor Components

$$\begin{aligned}\Phi_{00} = & -\frac{1}{2}e^{(2U-2k)}\frac{\partial}{\partial t}U^2 + e^{(2U-2k)}\frac{\partial}{\partial t}U\frac{\partial}{\partial\rho}U - \frac{1}{2}e^{(2U-2k)}\frac{\partial}{\partial\rho}U^2 + \frac{e^{(2U-2k)}\frac{\partial}{\partial t}W\frac{\partial}{\partial t}k}{2W} - \frac{e^{(2U-2k)}\frac{\partial}{\partial\rho}W\frac{\partial}{\partial t}k}{2W} \\ & - \frac{e^{(2U-2k)}\frac{\partial}{\partial t}W\frac{\partial}{\partial\rho}k}{2W} + \frac{e^{(2U-2k)}\frac{\partial}{\partial\rho}W\frac{\partial}{\partial\rho}k}{2W} - \frac{e^{(2U-2k)}\frac{\partial^2}{(\partial t)^2}W}{4W} + \frac{e^{(2U-2k)}\frac{\partial^2}{\partial t\partial\rho}W}{2W} - \frac{e^{(2U-2k)}\frac{\partial^2}{(\partial\rho)^2}W}{4W}\end{aligned}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = -\frac{1}{2}e^{(2U-2k)}\frac{\partial^2}{(\partial t)^2}U + \frac{1}{2}e^{(2U-2k)}\frac{\partial^2}{(\partial\rho)^2}U - \frac{e^{(2U-2k)}\frac{\partial}{\partial t}U\frac{\partial}{\partial t}W}{2W} + \frac{e^{(2U-2k)}\frac{\partial}{\partial\rho}U\frac{\partial}{\partial\rho}W}{2W} + \frac{e^{(2U-2k)}\frac{\partial^2}{(\partial t)^2}W}{4W} - \frac{e^{(2U-2k)}\frac{\partial^2}{(\partial\rho)^2}W}{4W}$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{1}{4}e^{(2U-2k)}\frac{\partial}{\partial t}U^2 + \frac{1}{4}e^{(2U-2k)}\frac{\partial}{\partial\rho}U^2 + \frac{1}{4}e^{(2U-2k)}\frac{\partial^2}{(\partial t)^2}U - \frac{1}{4}e^{(2U-2k)}\frac{\partial^2}{(\partial\rho)^2}U + \frac{e^{(2U-2k)}\frac{\partial}{\partial t}U\frac{\partial}{\partial t}W}{4W} - \frac{e^{(2U-2k)}\frac{\partial}{\partial\rho}U\frac{\partial}{\partial\rho}W}{4W} - \frac{1}{4}e^{(2U-2k)}\frac{\partial^2}{(\partial t)^2}k + \frac{1}{4}e^{(2U-2k)}\frac{\partial^2}{(\partial\rho)^2}k$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = -\frac{1}{2}e^{(2U-2k)}\frac{\partial^2}{(\partial t)^2}U + \frac{1}{2}e^{(2U-2k)}\frac{\partial^2}{(\partial\rho)^2}U - \frac{e^{(2U-2k)}\frac{\partial}{\partial t}U\frac{\partial}{\partial t}W}{2W} + \frac{e^{(2U-2k)}\frac{\partial}{\partial\rho}U\frac{\partial}{\partial\rho}W}{2W} + \frac{e^{(2U-2k)}\frac{\partial^2}{(\partial t)^2}W}{4W} - \frac{e^{(2U-2k)}\frac{\partial^2}{(\partial\rho)^2}W}{4W}$$

$$\Phi_{21} = 0$$

$$\begin{aligned}\Phi_{22} = & -\frac{1}{2}e^{(2U-2k)}\frac{\partial}{\partial t}U^2 - e^{(2U-2k)}\frac{\partial}{\partial t}U\frac{\partial}{\partial\rho}U - \frac{1}{2}e^{(2U-2k)}\frac{\partial}{\partial\rho}U^2 + \frac{e^{(2U-2k)}\frac{\partial}{\partial t}W\frac{\partial}{\partial t}k}{2W} + \frac{e^{(2U-2k)}\frac{\partial}{\partial\rho}W\frac{\partial}{\partial t}k}{2W} \\ & + \frac{e^{(2U-2k)}\frac{\partial}{\partial t}W\frac{\partial}{\partial\rho}k}{2W} + \frac{e^{(2U-2k)}\frac{\partial}{\partial\rho}W\frac{\partial}{\partial\rho}k}{2W} - \frac{e^{(2U-2k)}\frac{\partial^2}{(\partial t)^2}W}{4W} - \frac{e^{(2U-2k)}\frac{\partial^2}{\partial t\partial\rho}W}{2W} - \frac{e^{(2U-2k)}\frac{\partial^2}{(\partial\rho)^2}W}{4W}\end{aligned}$$

$$\begin{aligned}\Lambda = & \frac{1}{12} e^{(2U-2k)} \frac{\partial}{\partial t} U^2 - \frac{1}{12} e^{(2U-2k)} \frac{\partial}{\partial \rho} U^2 - \frac{1}{12} e^{(2U-2k)} \frac{\partial^2}{(\partial t)^2} U + \frac{1}{12} e^{(2U-2k)} \frac{\partial^2}{(\partial \rho)^2} U - \frac{e^{(2U-2k)} \frac{\partial}{\partial t} U \frac{\partial}{\partial t} W}{12 W} \\ & + \frac{e^{(2U-2k)} \frac{\partial}{\partial \rho} U \frac{\partial}{\partial \rho} W}{12 W} + \frac{1}{12} e^{(2U-2k)} \frac{\partial^2}{(\partial t)^2} k - \frac{1}{12} e^{(2U-2k)} \frac{\partial^2}{(\partial \rho)^2} k + \frac{e^{(2U-2k)} \frac{\partial^2}{(\partial t)^2} W}{12 W} - \frac{e^{(2U-2k)} \frac{\partial^2}{(\partial \rho)^2} W}{12 W}\end{aligned}$$

Weyl Tensor Components

$$\begin{aligned}\Psi_0 = & e^{(2U-2k)} \frac{\partial}{\partial t} U^2 - 2 e^{(2U-2k)} \frac{\partial}{\partial t} U \frac{\partial}{\partial \rho} U + e^{(2U-2k)} \frac{\partial}{\partial \rho} U^2 - e^{(2U-2k)} \frac{\partial}{\partial t} U \frac{\partial}{\partial t} k + e^{(2U-2k)} \frac{\partial}{\partial \rho} U \frac{\partial}{\partial t} k + e^{(2U-2k)} \frac{\partial}{\partial t} U \frac{\partial}{\partial \rho} k \\ & - e^{(2U-2k)} \frac{\partial}{\partial \rho} U \frac{\partial}{\partial \rho} k + \frac{1}{2} e^{(2U-2k)} \frac{\partial^2}{(\partial t)^2} U - e^{(2U-2k)} \frac{\partial^2}{\partial t \partial \rho} U + \frac{1}{2} e^{(2U-2k)} \frac{\partial^2}{(\partial \rho)^2} U + \frac{e^{(2U-2k)} \frac{\partial}{\partial t} W \frac{\partial}{\partial t} k}{2 W} \\ & - \frac{e^{(2U-2k)} \frac{\partial}{\partial \rho} W \frac{\partial}{\partial t} k}{2 W} - \frac{e^{(2U-2k)} \frac{\partial}{\partial t} W \frac{\partial}{\partial \rho} k}{2 W} + \frac{e^{(2U-2k)} \frac{\partial}{\partial \rho} W \frac{\partial}{\partial \rho} k}{2 W} - \frac{e^{(2U-2k)} \frac{\partial^2}{(\partial t)^2} W}{4 W} + \frac{e^{(2U-2k)} \frac{\partial^2}{\partial t \partial \rho} W}{2 W} - \frac{e^{(2U-2k)} \frac{\partial^2}{(\partial \rho)^2} W}{4 W}\end{aligned}$$

$$\Psi_1 = 0$$

$$\begin{aligned}\Psi_2 = & \frac{1}{3} e^{(2U-2k)} \frac{\partial}{\partial t} U^2 - \frac{1}{3} e^{(2U-2k)} \frac{\partial}{\partial \rho} U^2 + \frac{1}{6} e^{(2U-2k)} \frac{\partial^2}{(\partial t)^2} U - \frac{1}{6} e^{(2U-2k)} \frac{\partial^2}{(\partial \rho)^2} U - \frac{e^{(2U-2k)} \frac{\partial}{\partial t} U \frac{\partial}{\partial t} W}{3 W} \\ & + \frac{e^{(2U-2k)} \frac{\partial}{\partial \rho} U \frac{\partial}{\partial \rho} W}{3 W} - \frac{1}{6} e^{(2U-2k)} \frac{\partial^2}{(\partial t)^2} k + \frac{1}{6} e^{(2U-2k)} \frac{\partial^2}{(\partial \rho)^2} k + \frac{e^{(2U-2k)} \frac{\partial^2}{(\partial t)^2} W}{12 W} - \frac{e^{(2U-2k)} \frac{\partial^2}{(\partial \rho)^2} W}{12 W}\end{aligned}$$

$$\Psi_3 = 0$$

$$\begin{aligned}\Psi_4 = & e^{(2U-2k)} \frac{\partial}{\partial t} U^2 + 2 e^{(2U-2k)} \frac{\partial}{\partial t} U \frac{\partial}{\partial \rho} U + e^{(2U-2k)} \frac{\partial}{\partial \rho} U^2 - e^{(2U-2k)} \frac{\partial}{\partial t} U \frac{\partial}{\partial \rho} k - e^{(2U-2k)} \frac{\partial}{\partial \rho} U \frac{\partial}{\partial t} k - e^{(2U-2k)} \frac{\partial}{\partial t} U \frac{\partial}{\partial \rho} k \\ & - e^{(2U-2k)} \frac{\partial}{\partial \rho} U \frac{\partial}{\partial \rho} k + \frac{1}{2} e^{(2U-2k)} \frac{\partial^2}{(\partial t)^2} U + e^{(2U-2k)} \frac{\partial^2}{\partial t \partial \rho} U + \frac{1}{2} e^{(2U-2k)} \frac{\partial^2}{(\partial \rho)^2} U + \frac{e^{(2U-2k)} \frac{\partial}{\partial t} W \frac{\partial}{\partial t} k}{2 W} \\ & + \frac{e^{(2U-2k)} \frac{\partial}{\partial \rho} W \frac{\partial}{\partial t} k}{2 W} + \frac{e^{(2U-2k)} \frac{\partial}{\partial t} W \frac{\partial}{\partial \rho} k}{2 W} + \frac{e^{(2U-2k)} \frac{\partial}{\partial \rho} W \frac{\partial}{\partial \rho} k}{2 W} - \frac{e^{(2U-2k)} \frac{\partial^2}{(\partial t)^2} W}{4 W} - \frac{e^{(2U-2k)} \frac{\partial^2}{\partial t \partial \rho} W}{2 W} - \frac{e^{(2U-2k)} \frac{\partial^2}{(\partial \rho)^2} W}{4 W}\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"